

Juvenile Myoclonic Epilepsy: A 5-Year Prospective Study

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Summary: We made a long term prospective study of 66 patients with juvenile myoclonic epilepsy (JME). Prevalence was 10.2% among 672 patients with epilepsies. Sex distribution was equal. Sixty-three were not diagnosed on referral; JME was not initially recognized in the epilepsy clinic in 22. Clinical typical absence seizures were reported in 33.3%, myoclonic jerks in 97% and generalized tonic-clonic seizures (GTC) in 78.8% of the patients. Mean age ($\pm$ SD) at onset was 10.5 ± 3.4 years (range 5–16 years) for absence seizures, 15 ± 3.5 years (range 8–26 years) for myoclonic jerks, and 16 ± 3.5 years (9–28) for GTC. Absence predated myoclonic jerks by 3.9 ± 2.3 years (range 1–9 years) and GTC by 4.4 ± 2.7 years (range 1–8 years) in 14 (21.2%) patients who manifested all three types of seizure. Absence were never antedated by myoclonic jerks or GTC. Myoclonic jerks occurred on awakening in 87.5% of the patients. GTC occurred mainly on awakening, but other patients had nocturnal or diurnal GTC with no circadian distribution. Neurologic examination was normal for all patients except for tremor of the hands similar to essential tremor, noted in 35% of patients. Computed tomography (CT) brain scans were normal: 93% of patients had precipitating factors: sleep deprivation (89.5%), fatigue (73.7%), photosensitivity (36.8%; television and video games 8.8%), menstruation (24.1% of women), mental concentration (22.8%), and

stress (12.3%). Incidence of JME among siblings (13 of 41 examined families) implies an autosomal recessive mode of inheritance for this Arab population. EEGs were frequently normal in treated patients. At least one abnormal EEG was recorded in 56 (84.9%) patients. Abnormalities consisted mainly of generalized discharges of spike/double spike and/or polyspike and slow wave. Frequent multiple spikes and discharge fragmentations varied from 0.5- to 20-s duration (mean 6.8 s). Twenty (30.3%) had focal abnormalities, and 18 (27.3%) had photoconvulsive discharges. Eighty-eight percent of patients remained seizure-free for ≥ 3 years of follow-up. Effective treatment was achieved with valproate (VPA); control of myoclonic jerks was improved with clonazepam (CZP). CZP monotherapy did not consistently prevent GTC. Adding small doses of CZP with simultaneous reduction of VPA was the most effective and better tolerated form of medication, particularly in patients demonstrating an adverse reaction or requiring a large VPA dosage. VPA dosage was successfully reduced in 15 patients who were seizure-free for >2 years and had infrequent seizures before treatment, but 9 of 11 patients relapsed after VPA discontinuation. **Key Words:** Juvenile myoclonic epilepsy—Prevalence—Absence seizures—Generalized tonic-clonic seizures—Myoclonic jerks—Precipitating factors—Circadian distribution—Inheritance patterns—Valproate—Clonazepam.

Juvenile myoclonic epilepsy (JME) is a generalized, idiopathic epileptic syndrome with age-related onset with a prevalence of 5–11% among other epilepsies (Janz and Christian, 1957; Janz, 1969, 1985, 1989; Obeid and Panayiotopoulos, 1988; Commission, 1989; Grunewald et al., 1992). JME is characterized by myoclonic jerks, occurring mainly on awakening and often associated with generalized tonic-clonic seizures (GTC) and typical absence seizures in an age-related sequence. Seizure-precipitating factors, mainly sleep deprivation and

fatigue, are common. JME appears usually in the midteens and is probably a lifelong disorder (Janz, 1969, 1985, 1989; Asconapé and Penry, 1984; Delgado-Escueta and Enrile-Bacsal, 1984; Wolf, 1985; Obeid and Panayiotopoulos, 1988; Dreifuss, 1989; Panayiotopoulos et al., 1989a,b; Penry et al., 1989; Grunewald and Panayiotopoulos, 1993).

JME is an inherited disorder, but the pattern of transmission is controversial (Tsuboi and Christian, 1973; Panayiotopoulos and Obeid, 1989; Greenberg et al., 1988, 1990; Panayiotopoulos et al., 1991). Autosomal recessive (Panayiotopoulos and Obeid, 1989) and polygenic (Tsuboi and Christian, 1973) patterns of inheritance have been suggested but have not been accepted by other investigators (Greenberg et al., 1990). Recent reports indicated that JME may be tightly linked to the BF-HLA locus

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on chromosome 6 (Delgado-Escueta et al., 1989; Durner et al., 1991; Weissbecker et al., 1991), but this was not confirmed by other investigators (Whitehouse et al., 1992).

The EEG of untreated patients is usually abnormal, with generalized discharges described as fast 4- to 6-Hz polyspike-slow wave; one third of patients exhibit photoconvulsive responses (PCR) (Janz, 1985, 1989; Asconapé and Penry, 1984; Delgado-Escueta and Enrile-Bacsal, 1984; Wolf and Goosses, 1986). The EEG pattern of JME recently was more clearly identified (Panayiotopoulos et al., 1989a,b).

At least 80% of patients may become seizure-free with appropriate medication, but relapses are inevitable on treatment discontinuation (Covanis et al., 1982; Delgado-Escueta and Enrile-Bacsal, 1984; Obeid and Panayiotopoulos, 1988; Panayiotopoulos et al., 1989a; Penry et al., 1989; Christe, 1989; Janz, 1991).

Despite the distinctive features of JME and the significance of diagnosis for management, prognosis, and inheritance, JME is widely underdiagnosed (Delgado-Escueta and Enrile-Bacsal, 1984; Janz, 1985; Obeid and Panayiotopoulos, 1988; Penry et al., 1989; Panayiotopoulos et al., 1991; Anonymous, 1992; Grunewald et al., 1992). Recognition of the significance of JME in the English language neurologic literature, based on retrospective studies, is new.

We present our cumulative data from a 5-year prospective JME study. We know of no other similar study in which comprehensive clinical, genetic, EEG, video-EEG, radiologic, and therapeutic aspects of JME were studied prospectively. Results obtained in the course of our study regarding clinical and video-EEG manifestations of typical absence seizures, genetics, and factors of error in diagnosing JME were previously published (Panayiotopoulos and Obeid, 1989; Panayiotopoulos et al., 1989a,b, 1991) and are only briefly described herein.

PATIENTS AND METHODS

Patients

Our methods and patients were described previously (Panayiotopoulos et al., 1986, 1989a,b, 1991). An epilepsy clinic was established in King Khalid University Hospital, Riyadh, in 1983. Patients with seizures were referred from the hospital primary care physicians. Most patients were evaluated by more than 1 of 3 to 4 consultant neurologists. In view of the difficulty in eliciting the characteristic

myoclonic jerks of JME, we had adopted the policy of physically demonstrating to our patients and their relatives mild myoclonic jerks confined to the hands and inquiring about possible events associated with waking from sleep (particularly morning tremors and clumsiness). Similarly, in view of the often inconspicuous character of absence seizures, we consistently inquired about mild and transient events of impairment of consciousness. All patients diagnosed with JME were screened to confirm that they fulfilled the study inclusion and exclusion criteria. Patients who satisfied JME criteria were studied prospectively with a uniform protocol, recording seizures, precipitating factors, circadian distribution, and family history.

Inclusion criteria for JME were (a) unequivocal clinical evidence of generalized seizures with myoclonic jerks mainly on awakening; (b) no evidence of neurologic or intellectual deficit; (c) abnormal EEG in untreated patients with generalized spike and/or multiple spike-slow wave discharges (focal or background abnormalities were acceptable if other clinical and EEG criteria were strictly justified, and a normal EEG in treated patients was common); and (d) normal CT brain scan (if performed).

Exclusion criteria were (a) clinical and/or EEG evidence of myoclonic jerks secondary to brain hypoxia, metabolic diseases, or other structural brain abnormalities; (b) other epileptic syndromes such as eyelid myoclonia with absence seizures, pure forms of photosensitive epilepsy, or self-induced epilepsy; (c) generalized seizures or tonic-clonic and/or absence without firm documentation of myoclonic jerks; and (d) EEG abnormalities but no clinical evidence of any type of seizures.

Unlike Delgado-Escueta et al. (1989), we enrolled patients whose "petit mal seizures included characteristics of childhood absence, e.g., seizures in clusters of 20–100 per day and clinically photosensitive patients," provided that inclusion and exclusion criteria were justified. We also included 2 siblings with typical absence and no other form of seizures because of a strong family history of JME (mother, sister, and brother) indicating that these 2 patients were diagnosed before development of myoclonic jerks.

EEG

All patients underwent conventional EEGs, usually annually. Intermittent photic stimulation (IPS) was applied with combined flash and pattern stimulation (Panayiotopoulos et al., 1989a). Patients were tested without modification of current drug treatment.

Video-EEG studies were performed through a 16-channel Telefactor Modac 1, mainly in patients with absence seizures. Our methods of recording and evaluation were described previously (Panayiotopoulos et al., 1989a,b). EEG discharges were evaluated for duration, frequency, and morphology of the complexes.

Treatment

Most patients' seizures were not controlled with the antiepileptic drug (AED) being received at the time of referral. Usually the AED was phenytoin (PHT) or carbamazepine (CBZ); only a few patients were receiving valproate (VPA) (Panayiotopoulos et al., 1991).

We initiated treatment with VPA ~300–500 mg twice daily (b.i.d.) as a starting dose. Other AEDs were discontinued slowly at monthly decrements of 50 mg for PHT, 100 mg for CBZ, and 10–20 mg for phenobarbital (PB). VPA was increased at steps of 200–400 mg/week until seizure control was achieved or unacceptable adverse reactions developed. Patients with side effects from VPA (particularly tremors, falling hair, weight gain) received clonazepam (CZP) 0.5–3 mg/day as partial substitution for VPA. Monotherapy with CZP tried in a few patients at the initial stages of this study but was abandoned because some of the patients had or developed violent GTC (Obeid and Panayiotopoulos, 1989). Values are mean $\pm$ SD.

RESULTS

Prevalence of JME and of seizure type

Prevalence of JME and of seizure type is shown in Fig. 1 and Table 1. Sixty-six patients (10.2%) fulfilled the criteria for JME among 672 patients attending the epilepsy clinic: 22 (33.3%) had typical absence seizures, 64 (97.0%) had myoclonic jerks, 52 (78.8%) had GTC. It was exceptional for a patient to have only one type of seizure: 6 (9.1%) patients had myoclonic jerks alone, and 2 (3%) young patients of a JME family had absence seizures only. Fourteen patients (21.2%) had the triad of absence

seizures, myoclonic jerks, and GTC. Six (9.1%) patients had absence seizures and myoclonic jerks. Myoclonic jerks and GTC was the most common combination (38 patients, 57.6%).

Patient sex

Male and female patients were equally affected overall and throughout the subgroups with absence seizures, myoclonic jerks, and GTC (Table 2), but a female predominance of early onset absence seizures was observed, with a male predominance of late-onset absence seizures. Male patients had a 3.5-fold predominance of asymptomatic absence seizures.

Patient age and age at seizure onset

Mean age of patients on referral was 22.1 ± 7 years (range 8–40 years). Absence seizures had onset at 10.5 ± 3.4 years (range 5–16 years). In 8 of the 22 patients with absence seizures had onset before age 10 years (median 6 years). Myoclonic jerks began at age 15.0 ± 3.5 years (8–26 years), and GTC began at 16.0 ± 3.5 years (range 9–28 years).

Chronologic sequence of seizure type

Typical absence predated myoclonic jerks and GTC in all but 2 patients, who had simultaneous onset of absence seizures and myoclonic jerks at age 16 years (Fig. 2 and Table 1). Absence seizures predated myoclonic jerks by 3.4 ± 2.5 years (range 0–9 years). Myoclonic jerks took longer to appear in the 8 patients with early onset (4.9 ± 2.6 years) than in the 14 patients with late onset (2.6 ± 2.0 years) of absence seizures (Table 1).

In 14 patients with all three types of seizure, absence seizures began at age 10.3 ± 3.0 years (range 5–14 years), myoclonic jerks began at age 14.2 ± 2.5 years (range 9–18 years), and GTC began at age 14.7 ± 2.1 years (range 11–18 years). Absence seizures predated myoclonic jerks and GTC by 3.9 ± 2.3 years (range 1–9, median 5 years) and 4.4 ± 2.7 years (range 1–8, median 8 years), respectively. In these 14 patients, GTC occurred after a mean of 0.5 ± 1.6 years from onset of myoclonic jerks. Maximum time between onset of myoclonic jerks and GTC was 4 years.

In 38 patients with a combination of myoclonic jerks and GTC, jerks had onset at age 15.6 ± 3.9 years (range 8–26 years), and GTCs had onset at age 16.5 ± 3.8 years (range 9–28 years). Myoclonic jerks predated GTC by an average of 0.9 ± 3.8 years. GTC predated myoclonic jerks in 6, followed myoclonic jerks in 13, and appeared at the same age in 19 patients.

Myoclonic jerks alone occurred in 6 (9.1%) patients (4 female, 2 male). Age of onset was $15.2 \pm$

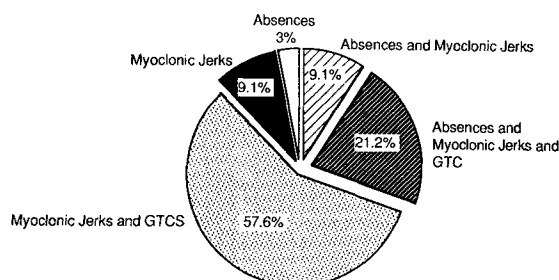


FIG. 1. Frequency of seizure type in juvenile myoclonic epilepsy. GTC, generalized tonic-clonic seizures.

TABLE 1. Age at onset of typical seizures, absence, myoclonic jerks, and GTC

Seizure type	n (%)	Onset (yr)					
		Absence		Myoclonic jerks		GTC	
		Mean $\pm$ SD	Range (yr)	Mean $\pm$ SD	Range (yr)	Mean $\pm$ SD	Range (yr)
Absence	22 (33.3)	10.5 $\pm$ 3.4	5–16				
Myoclonic jerks	64 (97.0)			15.0 $\pm$ 3.5	8–26		
GTC	52 (78.8)					16.0 $\pm$ 3.5	9–28
Absence only	2 (3.0)		6,13				
Myoclonic jerks only	6 (9.1)			15.2 $\pm$ 2.1	13–18		
Absence and myoclonic jerks	6 (9.1)	10.8 $\pm$ 4.0	5–16	12.9 $\pm$ 2.4	8–16		
Myoclonic jerks, absence, GTC	14 (21.2)	10.3 $\pm$ 3.0	5–14	14.2 $\pm$ 2.5	9–18	14.7 $\pm$ 2.1	11–18
Myoclonic jerks and GTC	38 (57.6)			15.6 $\pm$ 3.9	8–26	16.5 $\pm$ 3.8	9–28

GTC, generalized tonic-clonic seizures. Myoclonic jerks and GTC tended to appear earlier in patients with absence seizures.

2.1 years (range 13–18 years). Overall, the difference in years between onset of myoclonic jerks and GTC was 0.8 ± 3.4 years (range 6–18 years).

Circadian distribution

Myoclonic jerks

Myoclonic jerks in 56 (87.5%) patients occurred after they awoke from night sleep or a nap. In the remaining 8 patients (12.5%), myoclonic jerks were random during the day (7 patients) or were more frequent at night (1 patient).

GTC

The circadian distribution of GTC was studied in 16 patients. Seven had GTC on awakening only, and 4 had GTC mainly on awakening but also during sleep (3 patients) and randomly (1 patient). Two patients had nocturnal GTC only. The remaining 3 had diurnal GTC with no definite circadian distribution (in 1 of them, GTC were elicited only by photic stimuli).

Typical absence seizures

Typical absence seizures could occur at any time of the day, especially during excitation and mental concentration. No relationship to waking was noted, although circadian variation was not specifically studied.

Precipitating factors

Precipitating factors were detected in 53 (93.0%) of 57 patients who could provide reliable informa-

tion (Table 3). Sleep deprivation (51 patients) and fatigue (42 patients) were the most common precipitating factors. Photosensitivity was reported by 21 patients and was confirmed by EEG in 8. Conversely, 10 other patients with no clinical history of photosensitivity demonstrated photoconvulsive responses to IPS. Thus, 31 (54.4%) patients had clinical and/or laboratory evidence of photosensitivity (17 female, 14 male). Five patients also had a history of seizures precipitated by television and/or video games, 3 of whom demonstrated photosensitivity to IPS. Menstruation was considered a precipitating factor by 7 (24.1%) of 29 women. Mental concentration (13 patients, 22.8%) and stress, excitement, expectation, or frustration (as from an expected but unachieved goal in a football match, 7 patients) were other precipitating factors. One patient gave a clear history of attacks precipitated only by thinking and concentration, with no other precipitating factors involved. Frequently, a combination of precipitating factors was apparent, with patients reporting a higher incidence of seizures during school examination periods (sleep deprivation and fatigue, stress, mental concentration, frustration, or excitement).

Clinical manifestations of seizures

Myoclonic jerks

Myoclonic jerks were shocklike irregular and arrhythmic clonic movements of proximal and distal muscles, mainly of the upper extremities. They were of variable amplitude and force. They were sufficiently violent to cause falls in 40 (62.5%) of the 64 patients. In others, mild jerks occurred in the fingers, making the patient clumsy or prone to drop things unexpectedly. None of our patients complained of facial or eyelid myoclonus.

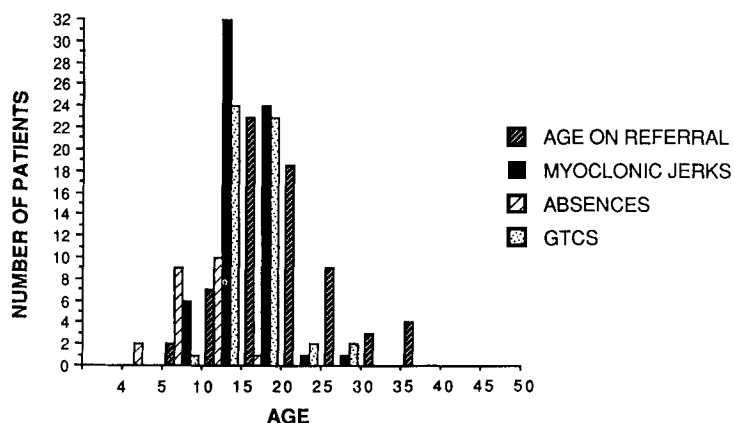
Myoclonic jerks were reported as bilateral by 49 (76.6%) of the patients. Fifteen patients (23.4%) described unilateral jerks: 11 (17.2%) on the right and

TABLE 2. Sex prevalence with seizure type

Seizure type	M	F	Ratio
Myoclonic jerks	33	31	1.1
GTC	27	25	1.1
Absence	12	10	1.2
Early-onset absence	3	5	0.6
Late-onset absence	9	5	1.8
Asymptomatic absence	7	2	3.5

GTC, generalized tonic-clonic seizures.

FIG. 2. Age at onset and referral in juvenile myoclonic epilepsy. GTC, generalized tonic-clonic seizures.



4 (6.2%) on the left. The hand dominance of the patients was not noted.

The characteristic history of myoclonic jerks was often difficult to elicit even in highly intelligent patients unless the concept of jerks was clearly understood by the patients and relatives. Diagnostic yield was improved by our policy of physically demonstrating mild myoclonic jerks of the hands, emphasizing resemblance to the normal phenomena of hypnagogic jerks and the close association to fatigue and sleep deprivation and specifically inquiring about morning tremulousness or clumsiness. We were surprised to detect jerks, denied by a few patients in a detailed interview, when we inquired about events preceding a GTC on awakening.

Typical absence seizures

The clinical severity of absence seizures appeared to be age related, as previously reported (Panayiotopoulos 1989a,b).

Early-onset absence seizures. Early onset absence seizures with severe impairment of consciousness were detected in 8 (12.1%) patients with onset before age 10 years.

Late-onset absence seizures. Fourteen (21.2%) patients had onset of absence seizures after age 10 years, but in only 3 were seizures sufficiently severe to cause ictal interruption of mental and physical activity. In the remaining 11 patients, absence

seizures tended to be inconspicuous, with little disturbance of consciousness. They were usually described as a momentary lapse of concentration and were not associated with myoclonic jerks and automatisms.

Asymptomatic absence seizures. Nine (13.7%) patients (7 male, 2 female) did not admit to absence seizures despite generalized spike-slow wave discharges lasting >3 s.

GTC. GTC were usually preceded by clusters of myoclonic jerks (clonic-tonic-clonic generalized seizures). In patients receiving only CZP, GTC were violent and were not preceded by myoclonic jerks.

Neurologic examination and CT brain scan

Neurologic examination was normal in all patients except for tremor of the hands similar to essential tremor in 23 (35%) patients. CT brain scans of 23 patients were normal.

EEG findings

Fifty-six (84.8%) patients had at least one abnormal EEG. Seven had normal and 3 had borderline EEG (mild excess of diffuse theta activity with occasional abortive paroxysmal theta waves). Paroxysmal abnormalities were recorded in 52 patients; 34 (51.5%) had spike/multiple spike-slow wave discharges, and 18 had brief generalized bursts of sharp theta activity with small spikes occasionally interspersed (Table 4).

TABLE 3. Precipitating factors

Factor	Alone or in combination, n (%)	Alone, n (%)
Sleep deprivation	51 (89.5)	10 (17.5)
Fatigue	42 (73.7)	2 (3.5)
Photic stimuli	21 (36.8)	3 (5.3)
Television/video games	5 (8.8)	1 (1.8)
Menstruation	7 (24.1)	
Concentration	13 (22.8)	1 (1.8)
Stress, expectation, others	7 (12.3)	
None	4 (7.0)	4 (7.0)

TABLE 4. EEG findings

Finding	n (%)
Normal	7 (10.6)
Abnormal	56 (84.8)
Borderline	3 (4.5)
Excessive diffuse slow activity	28 (42.4)
Paroxysmal activity	52 (78.8)
Spike/multiple spike slow wave discharge	34 (51.5)
Focal abnormality	20 (30.3)
Photoconvulsive response	18 (27.3)

Morphology and characteristics of discharges

Morphology and characteristics of the discharges were described previously (Panayiotopoulos et al., 1989a,b). The intradischarge frequency of the spike/multiple spike slow wave complexes was mainly 3–5 Hz, but individual complexes varied from 2 to 10 Hz. Fragmentations of the discharges and multiple spikes with a prominent compressed capital W appearance (Ws) were common and characteristic (Fig. 3). EEG discharges associated with myoclonic jerks consisted mainly of polyspikes (Fig. 4).

Focal abnormalities

Focal abnormalities (Figs. 3 and 5) were recorded in 20 (30.3%) patients. Ten had focal single spikes and spike-slow wave complexes. The other 10 patients showed focal slow waves consisting of sharp medium to high amplitude theta or delta waves, independently right or left in 9 and with unilateral preponderance in 6.

PCR

PCR (Fig. 5) were evoked in 18 (27.3%) patients; 3 had all types of seizures, 13 had myoclonic jerks

and GTC, 1 had myoclonic jerks and absence seizures, and 1 had only absence seizures.

Background abnormalities

Background abnormalities consisting of a moderately excessive theta activity and/or some slowing of the alpha rhythm were recorded in 28 (42.4%) patients.

Eyes-closed-related abnormalities

Eyes-closed-related abnormalities, i.e., abnormalities appearing and sustained during the period in which eyes are closed and which are eliminated or attenuated when eyes are open (Panayiotopoulos, 1974), were observed in 8 patients. Eye-closure-related abnormalities, i.e., transient EEG discharges occurring ≤ 1 –2 s after the eyes close but not sustained during the remaining period when eyes are either closed or open (Panayiotopoulos, 1974) occurred in 2. No effect was noted in 4 of 14 patients studied.

Hyperventilation

Hyperventilation accentuated the abnormalities in all patients.

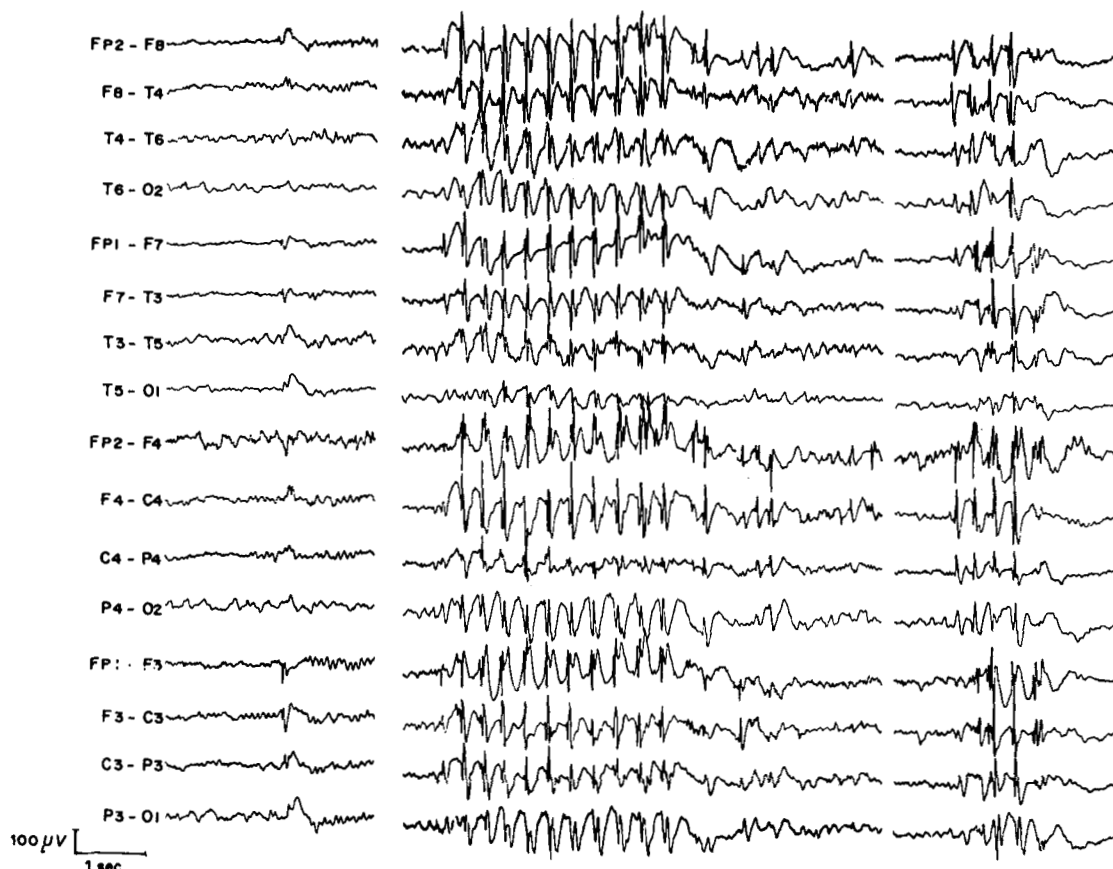


FIG. 3. EEG of a 22-year-old patient. There was no apparent clinical manifestations. Fragmentation of discharges, multiple spike complexes (Ws), asymmetries, and frequent focal abnormalities occurred independently right and left.

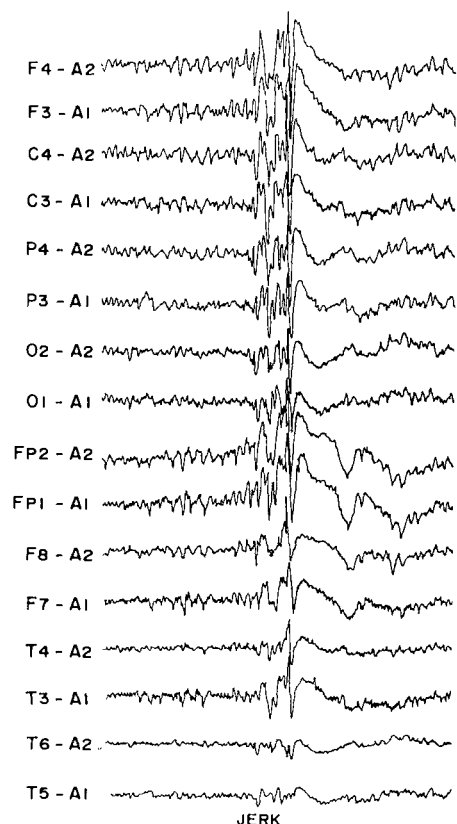


FIG. 4. EEG manifestations during a bilateral myoclonic jerk.

Family history

A reliable family history was obtained from 51 patients in 41 families (Table 5). A history of epilepsy was detected in 27 (65.9%) families: 15 (36%) families had at least 2 members affected by JME (Table 5). Siblings of 13 families had JME. Cousins of the probands and siblings of unaffected parents may have had JME. Parental and half-sibling involvement with JME were noted in 1 family each. In 3 of the families with at least 2 members with JME, other relatives were reported to have GTC (Table 5). Twelve other patients had relatives with seizures, but JME could not be confirmed: GTC in 7, symptomatic in 1, and febrile convulsions in 5 families of the probands.

Consanguinity was noted in 18 (43.9%) of 41 families. There were 13 (48.1%) consanguineous marriages among the 27 families with and among 5 (35.7%) of 14 without a family history of epilepsy. Consanguinity was noted in 6 (40%) of the 15 families with at least 2 JME members as compared with 7 (58.3%) of 12 families with seizures (no JME) in relatives of the probands.

Diagnosis on referral and in the epilepsy clinic

Sixty three (95.5%) of the patients were not correctly diagnosed on referral, and JME was not ini-

tially recognized in the epilepsy clinic in 22 (33.3%), as previously reported (Panayiotopoulos et al., 1991).

Treatment

Fifty patients had follow-up >3 years. Forty-three were receiving VPA either alone (29 patients) or in combination with other AEDs, mainly CZP. Patients with persistent myoclonic jerks despite optimal VPA dosage and those with VPA-related adverse reactions were treated by adding small doses of CZP (at night) and reducing VPA.

Forty-four (88%) patients became seizure-free, but in 10 occasional mild myoclonic jerking continued, particularly after sleep deprivation. In 7, seizure control was achieved only after CZP (1–4 mg daily) was combined with VPA. CZP monotherapy failed to prevent and in 1 patient may have precipitated GTC. However, 4 of 5 patients with CZP monotherapy were well controlled; 3 had a history of myoclonic jerks alone, and 1 had jerks with infrequent GTC.

Six patients continued to have seizures (GTC and myoclonic jerks) despite optimum treatment with VPA alone and/or in combination with other AEDs.

Discontinuation of treatment

In 15 patients who were seizure free for >2 years and had infrequent myoclonic jerks and GTC before treatment, VPA was reduced to 400 mg daily. There were no relapses. However, 1 of 4 patients who discontinued VPA treatment relapsed 12 months later another relapsed 16 months later. The other 2 remained seizure-free at 1 and 2 years of follow-up.

All 7 patients who discontinued medication relapsed: 4 developed GTC (1 and 3 weeks and 1 and 2 months after discontinuation), 1 had myoclonic status (after 2 weeks), 1 had absence status (after 3 weeks), and 1 had a relapse of violent myoclonic jerks.

DISCUSSION

Ours is the first 5-year prospective study of clinical, EEG, video-EEG characteristics and therapeutics in patients with JME.

Although a case of JME (a doctor's son) was substantially described by Herpin in 1867 (translated in English by Grunewald and Panayiotopoulos, 1993), the high prevalence of JME and its importance in genetics, prognosis, and treatment has only recently been appreciated. Despite well-defined clinical and EEG features, JME remains underdiagnosed not only by general physicians but also by experienced neurologists (Delgado-Escueta and Enrile-Bacsal, 1984; Janz, 1985; Obeid and Panayoto-

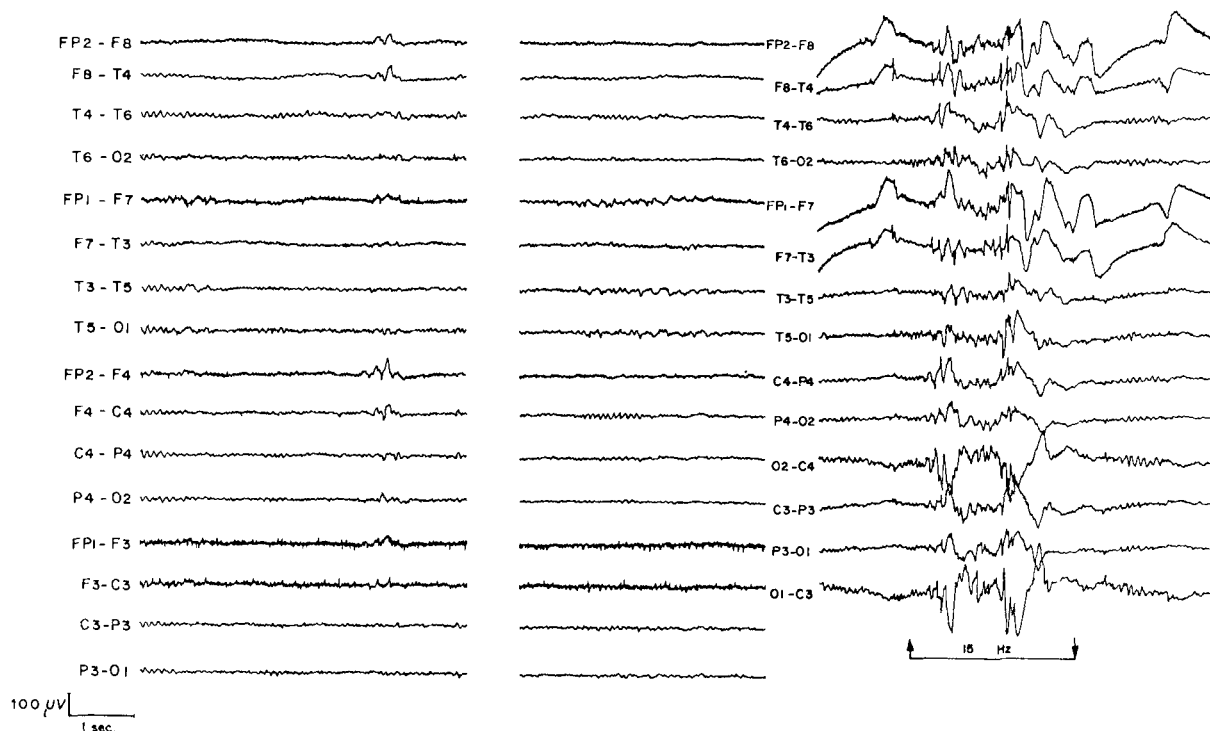


FIG. 5. EEG of a patient with focal abnormalities (**left and middle**) and photoconvulsive responses (**right**). Bursts of theta waves occurred on left and were of greater amplitude on the right (**left**). A unilateral long run of rhythmic theta activity is apparent (**middle**).

poulos, 1988; Grunewald et al., 1992; Panayiotopoulos et al., 1991). This was also our experience with 95.5% of patients not correctly diagnosed on referral and 33.3% of those not correctly diagnosed by the epilepsy clinic neurologists. The factors contributing to these errors and their consequences were detailed previously (Panayiotopoulos et al., 1991; Anonymous, 1992; Grunewald et al., 1992; Panayiotopoulos et al., 1992). The prevalence of JME in our epilepsy clinic was 10.2%, which is slightly higher than the 8.7% prevalence detected by us in England (Grunewald et al., 1992) and by other investigators worldwide (Janz, 1989; Obeid and Panayiotopoulos, 1989).

We confirmed that JME affects male and female patients equally (Janz and Christian, 1957; Janz, 1969; Tsuboi and Christian, 1973; Delgado-Escueta and Enrile-Bacsal, 1984; Goosses, 1984; Obeid and Panayiotopoulos, 1988) and demonstrated as well that sex distribution for seizure type is equal. However, early-onset absence seizures may be more common in girls, whereas late-onset and asymptomatic (phantom) absence seizures may occur more often in boys. This age-related sex preponderance may be of interest because more girls are affected by childhood absence epilepsy (Loiseau, 1992) and more boys are affected by juvenile (Wolf, 1992) and myoclonic absence epilepsy (Tassinari and Bureau,

1985), but this finding requires further confirmation because of the small numbers of these subgroups of our patients.

Mean age at onset of myoclonic jerks (15.5 years) in the present study compares well with previous reports of 14.6 years (Janz, 1989), 14.7 (Asconapé and Penry, 1984), 16.7 (Tsuboi, 1977), 13.6 (Delgado-Escueta and Enrile-Bacsal, 1984), and 12.3 (Clement and Wallace, 1990). It was unusual for a patient to have onset of myoclonic jerks after the second decade of life, and none of our patients developed myoclonic jerks after age 26 years, despite a claim that JME may begin in later life (Gram et al., 1988). None of our JME patients was aged >40 years while 67 other patients were aged 40–80 years (none having JME). This may indicate a decrease in seizure susceptibility after the fourth decade of life, contradicting suggestions that JME is a lifelong disorder (Dreifuss, 1989). Furthermore, the senior author (C.P.P.) of this report has examined >150 JME patients; only 1 man aged 65 years and 2 women aged 43 and 44 years had persisting seizures. These 3 patients all had frequent jerks, absence seizures, and GTCs; they were undiagnosed and unsuccessfully treated for 3–5 decades.

GTC usually began a few months to 1 year (maximum 6 years in this study) after onset of myoclonic jerks, as previously reported (Janz, 1989), but could

TABLE 5. Family size, consanguinity, and affected relatives of probands with JME

Patient/sex	Consanguineous marriage	Brothers/sisters (n)		Total	Seizure type/syndrome and relation (no.) with proband
C8/M	P	6	1	7	JME brother C27 ^a
C9/F	N	6	4	10	JME brother C10 ^a , female cousins (2) C24 ^a
C24/F	N	2	3	5	JME sister, cousins (2) C9 ^a , C10 ^a
C26/M	N	5	5	10	JME brother C42 ^a
C16/M	N	6	3	9	JME sisters C20 ^a , C51 ^a brothers C19 ^a , C50 ^a mother C31 ^a
C31/F	P	3	7	10	JME sons C16 ^a , C19 ^a , C50 ^a daughters C20 ^a , C51 ^a , sister
C41/M	N	6	5	11	JME brother, GTC uncle
C4/F	N	2	2	4	JME brother, GTC cousins (2)
C13/F	N/P	3	4	7	JME half-sister
C48/M	P(d)	4	1	5	JME brother, GTC cousins
C3/M	N	5	1	6	JME paternal half-sister C49 ^a
C49/F	P	7	2	9	JME nephew C3 ^a
C33/F	N	10	8	18	JME brother
C22/F	N	7	1	8	JME sister
C15/F	P	3	4	7	JME brother
C52/M	P	5	4	9	JME sister, brother
C53/M	N	3	2	5	JME brothers (2), sisters (1)
C34/M	P	1	1	2	JME cousin
C32/F	P(d)	4	5	9	GTC paternal (1) and maternal (1) brothers
C2/F	N	5	7	12	GTC paternal sister
C43/M	P(d)	4	3	7	GTC maternal sister
C12/M	P	6	3	9	GTC cousin (1)
C37/F	N	1	3	4	GTC maternal brother (1)
C44/F	P	4	1	5	GTC paternal sister, many febrile
C30/M	P	4	5	9	Febrile son (1)
C14/F	N	5	5	10	Febrile sister (1)
C6/F	P	6	6	12	Febrile sister (1)
C36/F	N	4	2	6	Febrile cousin (1)
C21/F	P	5	4	9	GTC paternal grandfather, febrile brother
C35/F	N	1	4	5	Symptomatic epilepsy, daughter
Total		133	106	239	
Mean		4	4	8	
SD		2	2	3	

JME, juvenile myoclonic epilepsy; N, no consanguineous marriage; P, consanguineous marriage of first cousins; NP, consanguinity with second wife only, who is the mother of the affected half-sister; P(d), distant relation; GTC, generalised tonic-clonic seizures, C, code number of patients included in this study. Ratio of male (23) to female (16) patients affected by JME in these families was 1.43; male (133) to female (106) ratio among sibships was 1.25.

^a Members included in this study.

predate myoclonic jerks. Absence was the first seizure type in one third of patients. Early-onset absence seizures predated myoclonic jerks and/or GTC, sometimes by as long as 9 years. Thus, children with typical absence seizures of JME may be diagnosed erroneously as having childhood absence epilepsy, a different epileptic syndrome (Panayiotopoulos et al., 1989b).

Myoclonic jerks alone, without GTC and absence seizures occurred in 9.1% of patients, a figure that compares well with previous reports of 7–17% (Asconapé and Penry, 1984; Janz, 1985, 1989; Penry et al., 1989). These percentage of patients may be underestimated, because myoclonic jerks may be mild, being misinterpreted as early morning clumsiness or nervousness. GTC were detected in 78.8%, as compared with 86–95% in previous studies (Tsuboi, 1977; Janz, 1989; Penry et al., 1989). These numbers are certainly influenced by early

recognition of JME and treatment. We noted clinical absence seizures in one third of patients; juvenile absence seizures were twice as common as childhood absence seizures. Typical absence seizures were previously reported in 14.6% (Janz, 1969), 21.5% (Tsuboi, 1977), and 39.5% (Delgado-Escueta and Enrile-Bacsal, 1984) of patients with JME. A characteristic clinical and EEG pattern of typical absence seizures in JME was reported previously (Panayiotopoulos et al., 1989a,b, 1992).

Study of the circadian distribution of seizures confirmed that myoclonic jerks occur on awakening (Janz, 1985, 1989), but GTC previously described as occurring only on awakening (Janz, 1989) were also nocturnal in 20% of our patients. These findings are consistent with a sleep study of JME (Touchon, 1982). CZP, and probably other AEDs, affected the circadian distribution of GTC, which may become nocturnal (Obeid and Panayiotopoulos, 1989).

The most common seizure-precipitating factors were sleep deprivation and fatigue. This finding is at variance with the results of Penry et al. (1989), who reported that seizures were exacerbated in all their 50 patients by stress (100%), strong emotions (96%), and chronic anxiety (70%), but less by sleep deprivation (42%), menses (15%), and alcohol consumption (30%). Photosensitivity is common in JME, although reported incidence varies, probably owing to AED differences, IPS methods, and patient age (Jeavons et al., 1980; Delgado-Escueta and Enrile-Bacsal, 1984; Wolf and Goosses, 1986; Janz, 1989; Penry et al., 1989). Eighteen patients had PCR, and 13 others reported sensitivity to light. A slight preponderance among females was noted, but we could not confirm the opinion of Janz (1989) that all female patients with only myoclonic jerks are photosensitive. In 5 patients, seizures were evoked by television and/or video games. The first case of JME and video game-induced seizures probably was reported by Dahlquist et al. (1983) of the Mayo clinic, but JME was not recognized and the patient was described as having "light-sensitive epilepsy" despite "frequent generalized spike and slow-wave complexes and myoclonus after arousal from sleep." Patients with JME may be particularly vulnerable to video games because of accumulated precipitating factors (photosensitivity, sleep deprivation, excitement and frustration, strong emotions); patients should be appropriately warned of the risks (Ferrie et al., 1994).

The neurologic and mental state of the patients was normal except for a high incidence of essential tremor, not previously described, which could not be attributed to medication being received before VPA treatment was initiated.

The phenotypic expression of JME in this Arab population does not appear to be different from that of JME of Greek and British patients studied by one of us (C.P.P.). Four of 18 patients with JME studied in England also had essential tremor (Grunewald et al., 1992).

The family history of the 66 patients we report is similar to that of a smaller sample of patients (who all had the triad of generalized seizures, i.e., absence seizures, myoclonic jerks, and GTC) reported previously and confirms the expressed view that JME may be an autosomal recessive disorder in this population in Saudi Arabia (Panayiotopoulos and Obeid, 1989). However, different views have been expressed (Greenberg et al., 1990). Whether the genetics of this Arab population are different from those of other JME populations requires further investigation.

Our study confirmed that VPA is the preferable

AED for treatment of JME and that treatment should not be discontinued even after patients are seizure-free for a long time (Covanis et al., 1982; Christe, 1989; Dreifuss, 1989; Obeid and Panayiotopoulos, 1989; Penry et al., 1989; Janz, 1991). Relapses occurred, with instances of status epilepticus, within weeks or months in patients who voluntarily discontinued AED. Patients with myoclonic jerks had a high incidence of relapse in a recent study of AED discontinuation in patients with seizures on remission (Medical Research Council, 1991). Our study also demonstrated that reducing AEDs in patients may be safe with infrequent seizures, but relapses are likely to occur months or even years after discontinuation of medication.

Patients with violent myoclonic jerks or adverse reactions to VPA benefited from small adjunctive doses of CZP (1–2 mg at night). However, CZP monotherapy did not prevent and possibly precipitated GTC in some patients. Patients were also deprived of the warning effect of myoclonic jerks (Obeid and Panayiotopoulos, 1989) that often herald impending GTC. In patients with myoclonic jerks alone, CZP monotherapy may provide control of seizures.

REFERENCES

- Anonymous. Diagnosing juvenile myoclonic epilepsy [Editorial]. *Lancet* 1992;340:759–60.
- Asconapé J, Penry K. Some clinical and EEG aspects of benign juvenile myoclonic epilepsy. *Epilepsia* 1984;25:108–14.
- Medical Research Council. Antiepileptic drug withdrawal in patients in remission. Randomised study of antiepileptic drug withdrawal in patients in remission. *Lancet* 1991;337:1175–80.
- Christe W. Valproate in juvenile myoclonic epilepsy. In: Chadwick D, ed. *Fourth international symposium on sodium valproate and epilepsy*. International Congress and Symposium Series. London: Royal Society of Medicine Services, 1989:91–4.
- Clement MJ, Wallace SJ. Juvenile myoclonic epilepsy. *Arch Dis Child* 1990;63:1049–53.
- Commission on Classification and Terminology of the International League Against Epilepsy. Proposal for classification of epilepsies and epileptic syndromes. *Epilepsia* 1989;30:389–99.
- Covanis A, Gupta AK, Jeavons PM. Sodium valproate: monotherapy and polytherapy. *Epilepsia* 1982;23:693–720.
- Dahlquist NR, Mellinger JF, Klass DW. Hazards of video games in patients with light-sensitive epilepsy. *JAMA* 1983;249:776–7.
- Delgado-Escueta AV, Enrile-Bacsal F. Juvenile myoclonic epilepsy of Janz. *Neurology* 1984;34:285–94.
- Delgado-Escueta AV, Greenberg DA, Treiman L, et al. Mapping the gene for juvenile myoclonic epilepsy. *Epilepsia* 1989;30 (suppl 4):S8–18.
- Dreifuss FE. Juvenile myoclonic epilepsy: characteristics of a primary generalised epilepsy. *Epilepsia* 1989;30(suppl 4):S1–7.
- Durner M, Sander T, Greenberg DA, Johnson K, Beck-Managetta G, Janz D. Localization of idiopathic generalized epilepsy on chromosome 6p in families of juvenile myoclonic epilepsy patients. *Neurology* 1991;41:1651–5.

- Ferrie CD, De Marco P, Grünwald R, Giannacodimos S, Panayiotopoulos CP. Video-game induced seizures. *J Neurol Neurosurg Psychiatry* 1994 (in press).
- Gram L, Alving J, Sagil JC, Dam M. Juvenile myoclonic epilepsy in unexpected age groups. *Epilepsy Res* 1988;2:137-40.
- Goosses R. Die beziehung der Fotosensibilität zu den verschiedenen epileptischen Syndromen. Medical Dissertation, Berlin: Freie Universität, 1984.
- Greenberg DA, Delgado-Escueta AV, Widelitz H, et al. Juvenile myoclonic epilepsy may be linked to the BF and HLA loci human chromosome. *Am J Med Genet* 1988;31:185-92.
- Greenberg DA, Durner M, Delgado-Escueta AV, Janz D. Is juvenile myoclonic epilepsy an autosomal recessive disease? *Ann Neurol* 1990;28:110-1.
- Grünwald R, Chroni E, Panayiotopoulos CP. Delayed diagnosis of juvenile myoclonic epilepsy. *J Neurol Neurosurg Psychiatry* 1992;55:497-9.
- Grünwald R, Panayiotopoulos CP. Juvenile myoclonic epilepsy: A review. *Arch Neurol* 1993;50:594-8.
- Herpin TH. *Des acces incomplets d'épilepsie*. Paris: J.B. Baillière et fils, 1867.
- Janz D. *Die Epilepsien: Spezielle Pathologie und Therapie*. Stuttgart: Thieme, 1969.
- Janz D. Epilepsy with impulsive petit mal (juvenile myoclonic epilepsy). *Acta Neurol Scand* 1985;72:449-559.
- Janz D (1989) Juvenile myoclonic epilepsy. Epilepsy with impulsive petit mal. *Cleve Clin J Med* 1989;56(suppl I):S23-33.
- Janz D. Remission and relapse in juvenile myoclonic epilepsy. *Boll Lega It Epil* 1991;76:19-28.
- Janz D, Christian W. Impulsive petit-mal. *Dtsch Z Nervenheilkd* 1957;176:346-86.
- Loiseau P. Childhood absence epilepsy. In: Roger J, Bureau M, Dravet C, Dreifuss FE, Perret A, Wolf P, eds. *Epileptic syndromes in infancy, childhood and adolescence*. London: John Libbey, 1992.
- Obeid T, Panayiotopoulos CP. Juvenile myoclonic epilepsy: a study in Saudi Arabia. *Epilepsia* 1988;29:280-2.
- Obeid T, Panayiotopoulos CP. Clonazepam in juvenile myoclonic epilepsy. *Epilepsia* 1989;30:603-6.
- Panayiotopoulos CP. Effectiveness of photic stimulation on various eye-states in photosensitive epilepsy. *J Neurol Sci* 1974;24:95-107.
- Panayiotopoulos CP. Epilepsies [Editorial]. *Ann Saudi Med* 1986;6:81-2.
- Panayiotopoulos CP, Obeid T. Juvenile myoclonic epilepsy: an autosomal recessive disease. *Ann Neurol* 1989;25:440-3.
- Panayiotopoulos CP, Obeid T, Waheed G. Absences in juvenile myoclonic epilepsy: a clinical and video-electroencephalographic study. *Ann Neurol* 1989a;25:391-7.
- Panayiotopoulos CP, Obeid T, Waheed G. Differentiation of typical absence seizures in epileptic syndromes. A video EEG study of 224 seizures in 20 patients. *Brain* 1989b;112:1039-56.
- Panayiotopoulos CP, Tahan R, Obeid T. Juvenile myoclonic epilepsy: factors of error involved in the diagnosis and treatment. *Epilepsia* 1991;32:672-6.
- Panayiotopoulos CP, Chroni E, Dascalopoulos C, Baker A, Rowlinson S, Welsh P. Typical absences in adults: a clinical, EEG, video-EEG study and syndromic/diagnostic considerations. *J Neurol Neurosurg Psychiatry* 1992;55:1002-8.
- Penry JK, Dean JC, Riela AR. Juvenile myoclonic epilepsy: long-term response to therapy. *Epilepsia* 1989;30(suppl 4):S19-23.
- Tassinari CA, Bureau M. Epilepsy with myoclonic absences. In: Roger J, Bureau M, Dravet C, Dreifuss FE, Perret A, Wolf P, eds. *Epileptic syndromes in infancy, childhood and adolescence*. London: John Libbey, 1992:121-36.
- Touche J. Effect of awakening on epileptic activity in primary generalized myoclonic epilepsy. In: Sterman MB, Shouse MN, Passouant P, eds. *Sleep and epilepsy*. New York: Academic Press, 1982:239-48.
- Tsuboi T. *Primary generalised epilepsy with sporadic myoclonias of myoclonic petit mal type*. Stuttgart: Thieme, 1977.
- Tsuboi T, Christian W. On the genetics of the primary generalised epilepsy with sporadic myoclonus of impulsive petit mal type. *Hum Genet* 1973;19:155-82.
- Weissbecker KA, Durner M, Janz D, Scaramelli A, Sparkes RS, Spence MA. Confirmation of linkage between juvenile myoclonic epilepsy locus and the HLA region on chromosome 6. *Am J Med Genet* 1991;38:32-6.
- Whitehouse WP, Rees M, Sundqvist A, Gardiner RM. Linkage analysis for JME: exclusion data for chromosome 6p including HLA. *Seizure* 1992;1(suppl A):P3/20.
- Wolf P. Juvenile myoclonic epilepsy. In: Roger J, Bureau M, Dravet C, Dreifuss FE, Perret A, Wolf P, eds. *Epileptic syndromes in infancy, childhood and adolescence*. London: John Libbey, 1992.
- Wolf P, Goosses R. Relationship of photosensitivity to epileptic syndromes. *J Neurology, Neurosurg Psychiatry* 1986;49:1386-91.

RÉSUMÉ

Les auteurs ont entrepris une étude prospective à long terme de 66 patients présentant une épilepsie myoclonique juvénile (EMJ). La prévalence était de 10.2% parmi 672 patients présentant une épilepsie. La distribution entre les sexes était égale. Soixante trois patients n'étaient pas diagnostiqués au moment de leur premier contact; l'épilepsie myoclonique juvénile n'avait pas été reconnue initialement dans la consultation épileptologique chez 22 patients. Les absences cliniques étaient reportées chez 33.3%, des secousses myocloniques chez 97% et des crises tonico-cloniques chez 78.8% des patients. L'âge moyen du début était de 10.5 ± 3.4 ans ($\pm$ SD, limite 5 à 16) pour les absences, 15 ± 3.5 (8 à 26) pour les secousses myocloniques et 16 ± 3.5 (9 à 28) pour les crises généralisées tonico-cloniques. Les absences précédaient les myoclonies de 3.4 ± 2.3 années (de 1 à 9) et les crises généralisées tonico-cloniques de 4.4 ± 2.7 années (1 à 8) chez les 14 patients (21.2%) qui présentaient les trois types de crises. Les absences n'ont jamais été précédées par des secousses myocloniques ou par des crises généralisées tonico-cloniques. Les secousses myocloniques étaient constatées au réveil chez 87.5% des patients. Les crises généralisées tonico-cloniques survenaient principalement au réveil, mais d'autres patients avaient des crises nocturnes ou diurnes sans distribution circadienne. L'examen neurologique était normal chez tous les patients, en dehors d'un tremblement des mains semblable au tremblement essentiel, constaté chez 35% des patients. Les scanners cérébraux ont toujours été normaux. Les facteurs déclenchant des crises ont été constatés chez 93% des patients: privation de sommeil chez 89.5%, fatigue chez 73.7%, photosensibilité chez 36.8% (télévision et jeux vidéo 8.8%), menstruation chez 24.1% des femmes, concentration mentale chez 22.8%, stress chez 12.3%. L'incidence de l'EMJ dans les fratries (13 des 41 familles examinées) implique un mode de transmission autosomique récessif dans cette population d'origine arabe. Les EEG étaient souvent normaux chez les patients traités. Au moins un EEG anormal a été enregistré chez 56 patients (84.9%). Les anomalies étaient représentées surtout par des décharges généralisées de pointes, doubles pointes ou poly pointes-ondes lentes. La durée des pointes multiples et des fragmentations des décharges variaient de 0.5 à 20 secondes (en moyenne 6,8 secondes). Les anomalies focales ont été constatées chez 20 patients (30.3%) et des décharges photoconvulsives chez 18 patients (27.3%). Quatre vingthuit pour cent des patients sont restés sans crise pendant 3 ans ou plus de suivi. Le traitement efficace a utilisé le valproate (VPA). Le contrôle des secousses myocloniques a été amélioré avec le clonazépam (CZP). Une monothérapie par CZP n'a pas permis de prévenir de façon constante les crises généralisées tonico-clonique. Une substitution partielle utilisant le valproate, avec de petites doses de CZP, a été le traitement le plus efficace et le mieux toléré, particulièrement chez les patients présentant des effets secondaires ou nécessitant de fortes doses de VPA. La dose de VPA a été réduite avec succès chez 14 patients qui n'avaient plus de crises depuis plus

de 2 ans et qui n'avaient que de rares crises avant le traitement. Cependant, 9 patients sur 11 ont récidivé après l'arrêt du VPA.

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RESUMEN

Los autores han llevado a cabo un estudio prospectivo a largo plazo de 66 enfermos con Epilepsia Mioclónica Juvenil (JME). La prevalencia fue de 10.2% en 672 casos con epilepsia. La distribución entre sexos fue equivalente. Sesenta y tres casos no habían sido diagnosticados por quienes enviaban los enfermos y la JME no fue reconocida inicialmente en la clínica de epilepsia en 22 sujetos. Los típicos ataques de ausencia clínica aparecieron en 33.3%, los movimientos mioclónicos en 97% y los ataques generalizados tónico-clónicos (GTCs) en 78.8% de los enfermos. La edad media ($\pm$ SD) al comienzo fue de 10.5 ± 3.4 (rango 5–16) años para las ausencias, 15 ± 3.5 (8–26) para los movimientos mioclónicos y 16 ± 3.5 (9–28) para los GTCs. Las ausencias precedieron a los movimientos mioclónicos en 3.4 ± 2.3 años (rango 1–9) y en 4.4 ± 2.7 a los GTCs (rango 1–8) y en 14 años (21.2%) en enfermos que presentaban los tres tipos de ataques. Las ausencias nunca fueron precedidas de ataques mioclónicos o GTCs. Las mioclonías se presentaron durante la vigilia en 87.5% de los enfermos. Los GTCs predominaron durante la vigilia pero otros enfermos tenían GTCs nocturnos o diurnos sin distribución circadiana. La exploración neurológica fue normal en todos los pacientes con la excepción de temblor en las manos semejante al temblor esencial que apareció en el 35% de los enfermos. El CT de cerebro fue normal. En el 93% de los

enfermos se encontraron factores precipitantes: la privación del sueño en 89.5%, la fatiga en 73.7%, la fotosensibilidad en 36.8% (televisión y juegos de video en 8.8%), la menstruación en el 24.1% de las mujeres, el retraso mental en el 22.8% y el stress en el 12.3%. La incidencia de los JME entre hermanos (13 en 41 de las familias examinadas) sugiere un modo autosómico recesivo de herencia para esta población árabe. Los EEGs fueron frecuentemente normales en los enfermos tratados. Al menos un EEG anormal fue registrado en 56 enfermos (84.9%). Las anomalías consistieron, generalmente, en descargas generalizadas de punta/doble punta y/o polipuntas y ondas lentas. La frecuencia de las múltiples puntas y la fragmentación de las descargas variaron de 0.5–20 s (promedio 6.8 s) de duración. Las anomalías focales se observaron en 20 (30.3%) y las descargas fotoconvulsivas en 18 (27.3%) pacientes. El 88% de los enfermos permanecieron libres de ataques durante 3 o más años de seguimiento. La eficacia del tratamiento se consiguió con valproato (VPA) que controló los movimientos mioclónicos y mejoró con clonazepam (CZP). La monoterapia con CZP no evitó de modo consistente las GTCs. La sustitución parcial de VPA por dosis bajas de CZP fue la medicación más eficaz y mejor tolerada particularmente en enfermos que demostraban reacciones adversas o requerían dosis altas de VPA. La reducción de las dosis de VPA fue posible en 14 enfermos que estaban libres de ataques durante más de 2 años y habían tenido ataques infrecuentes antes del tratamiento. Sin embargo 9 de 11 pacientes volvieron a tener episodios al interrumpir el VPA.

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